卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0801 30-B003701S-T0801

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 08 卷 第 01 册
Structure SPECIALITY VOLUME BOOK

卷册名称 极1阀厅及防火墙基础施工图
Volume name foundation construction for pole 1 valve hall and firewall

批 准 APPROVED BY	2017.11.27
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	王轩 2017.11.27
图 纸 6 张 DRAWING NOS.	打印件 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0801-001	极1阀厅及防火墙基础施工图设计总说明 General specification of foundations for pole 1 valve hall and firewall	A	1		
2	30-B003701S-T0801-002	极1阀厅及防火墙承台布置图 plan layout of pile cap for pole 1 valve hall and firewall	A	1		
3	30-B003701S-T0801-003	极1阀厅±0.000m结构布置图 ±0.000m structure layout for pole 1 valve hall and firewall	A	1		
4	30-B003701S-T0801-004	极1阀厅及防火墙承台详图(1) Detail drawing of pile cap for pole 1 valve hall and firewall(1)	A	1		
5	30-B003701S-T0801-005	极1阀厅及防火墙承台详图(2) Detail drawing of pile cap for pole 1 valve hall and firewall(1)	A	1		
6	30-B003701S-T0801-006	极1阀厅及防火墙承台详图(3) Detail drawing of pile cap for pole 1 valve hall and firewall(1)	A	1		

注:

基础施工图设计说明

GENERAL SPECIFICATION OF FOUNDATIONS

1 如无特别说明，本工程图纸所注尺的长度单位为毫米（mm），标高单位为米（m）。

1 Unless otherwise specified, the length of the drawings is in millimeters (mm) and the elevation is in meters (m).

2 ±0.000相当于绝对标高193.90（EGM2008高程系统），本册图纸中所注标高均为相对标高。

2 ±0.000 is equal to the absolute elevation of 193.90 (geodestic datum system EGM2008). The elevations in drawings are adopt to the relative elevation.

3 工程概况

3 Project Introduction

建筑层数：地上1层　　Building floors：1 floors on the ground

建筑平面尺寸（m）：82x53（长x宽）　　Building plan size(m)：82x53(Long x wide)

建筑高度（m）：35.30　　Bulding height(m)：35.3

地基基础形式：桩基础　　Foundation type：Pile foundation

上部结构形式：混凝土剪力墙+钢框架　　Structure type：Concrete shear wall+ Steel frame

场地土类型：S_D　　Soil profile type.：S_D

抗震分区：ZONE 2A　　Seismic Zone：ZONE 2A

基本风速：44.8m/s　　Basic Wind Speed：44.8m/s

暴露类别：C　　Exposure category：C

4 材料

4 Material

4.1 混凝土

4.1 Concrete

基础及基础连系梁：fc'=28MPa，垫层：fc'=21MPa，框架柱、梁及剪力墙：fc'=28MPa。

Foundations and connecting beams of the foundation ：fc'=28MPa

Cushions ：fc'=21MPa

Frame Beams, Beams and Shear Walls ：fc'=28MPa

4.2 钢筋

4.2 Steel bars

Φ表示 Grade40(fy=280MPa)钢筋，⚤表示 Grade60(fy=420MPa)钢筋，钢筋应符合标准ASTM A615 or ASTM A706的规定。

Φ represents the steel reinforcement of Grade40(fy=280MPa) and ⚤ represents Grade60(fy=420MPa). Steel bars should conform to ASTM A615 or ASTM A706.

4.3 混凝土耐久性

4.3 Concrete durability requirements

4.3.1 ±0.000以下混凝土暴露类别及等级为S1。

4.3.1 Exposure categories and classes of concrete under ±0.000 is S1.

4.3.2 根据暴露类别及等级，混凝土配合物应符合 ACI 318-14表19.3.2.1中所列的规定。

4.3.2 Based on the exposure classes, concrete mixtures shall conform to the most restrictive requirements in ACI 318-14 Table 19.3.2.1.

5 钢筋混凝土保护层厚度

5 Concrete cover requirements

钢筋混凝土保护层厚度指最外层钢筋外边缘至混凝土表面的距离。

The thickness of concrete cover for reinforcements is the distance from the outer edge of the outermost reinforcement to the surface of concrete.

混凝土最小保护层厚度表(mm)

Specified concrete cover for concrete members(mm)

混凝土暴露等级 Concrete exposure	构件 Member	钢筋 Reinforcement	保护层厚度 Specified cover
混凝土浇筑在地基上并永远处在地基土环境中 Cast against and permanently in contact with ground	所有 All	所有 All	77
混凝土处在露天环境或地基土环境中 Exposed to weather or in contact with ground	所有 All	No.19~No.57	51
		≤No.16	40
混凝土不处在露天环境或地基土环境中 Not exposed to weather or in contact with ground	板、墙 Slabs, Walls	No.43、No.57	40
		≤No.36	20
	梁、柱 Beams,Columns	主筋、箍筋及拉筋 Primary reinforcement, stirrups, ties, spirals and hoops	40

桩承台底面钢筋混凝土保护层厚度：100mm

The thickness of concrete cover for reinforcements in bottom of the pile cap is 100mm.

6 钢筋的构造

6 Reinforcement details

6.1 钢筋净距

6.1 Minimum spacing of reinforcement

6.1.1 对于平行的水平钢筋的净距取25.4mm、d_b(钢筋直径)和4/3d_{agg}(骨料最大尺寸)的最大值。

6.1.1 For parallel reinforcement in a horizontal layer, clear spacing shall be at least the greatest of 25.4mm, d_b(bar diameter), and 4/3 d_{agg} (nominal maximum size of coarse aggregate).

6.1.2 当平行的水平钢筋多层布置时，上下层钢筋应对齐，层净距不小于25.4mm。

6.1.2 For parallel reinforcement placed in two or more horizontal layers, reinforcement in the upper layers shall be placed directly above reinforcement in the bottom layer with a clear spacing between layers of at least 25.4mm.

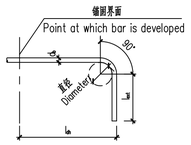
6.1.3 对于柱、基座、支柱和墙边界中的纵筋，钢筋净距取38mm，1.5d_b and 4/3 d_{agg}的最大值。

6.1.3 For longitudinal reinforcement in columns, pedestals, struts, and boundary elements in walls, clear spacing between bars shall be at least the greatest of 38mm, 1.5d_b and 4/3 d_{agg}.

6.2 受拉钢筋的弯钩构造

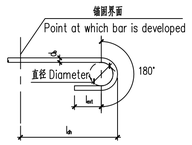
6.2 Standard hook geometry for development of deformed bars in tension

种类 Type	钢筋大小 Bar size	最小内弯直径 Minimum inside bend diameter	弯钩直段长度 l_{ext} Straight extension l_{ext}
90度弯钩 90-degree hook	No.10~No.25	$6d_b$	$12d_b$
	No.29~No.36	$8d_b$	
	No.43, No.57	$10d_b$	
180度弯钩 180-degree hook	No.10~No.25	$6d_b$	$4d_b$ 和 64mm取大值 Greater of $4d_b$ and 64mm
	No.29~No.36	$8d_b$	
	No.43, No.57	$10d_b$	



纵筋90度弯钩

90-degree hook of deformed bars



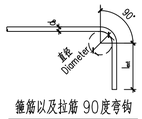
纵筋180度弯钩

180-degree hook of deformed bars

6.3 箍筋以及拉筋的弯钩构造

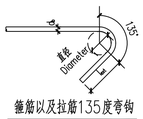
6.3 Standard hook geometry for stirrups, ties, and hoops

种类 Type	钢筋大小 Bar size	最小内弯直径 Minimum inside bend diameter	弯钩直段长度 l_{ext} Straight extension l_{ext}
90度弯钩 90-degree hook	No.10~No.16	$4d_b$	$6d_b$ 和 77mm取大值 Greater of $6d_b$ and 77mm
	No.19~No.25	$6d_b$	$12d_b$
135度弯钩 135-degree hook	No.10~No.16	$4d_b$	$6d_b$ 和 77mm取大值 Greater of $6d_b$ and 77mm
	No.19~No.25	$6d_b$	
180度弯钩 180-degree hook	No.10~No.16	$4d_b$	$4d_b$ 和 64mm取大值 Greater of $4d_b$ and 64mm
	No.19~No.25	$6d_b$	



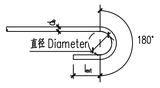
箍筋以及拉筋 90度弯钩

90-degree hook of stirrups, ties, and hoops



箍筋以及拉筋135度弯钩

135-degree hook of stirrups, ties, and hoops



箍筋以及拉筋180度弯钩

180-degree hook of stirrups, ties, and hoops

6.4 对于抗震设计的箍筋以及拉筋，其弯钩锚固应符合以下(a)和(b)的规定：

6.4 Seismic hooks used to anchor stirrups, ties, hoops, and crossties should be in accordance with (a) and (b):

(a) 环箍弯钩最小90度，其它箍筋最小135度。

(a) Minimum bend of 90 degrees for circular hoops and 135 degrees for all other hoops.

(b) 弯钩应与纵筋钩住，弯钩直段应弯向箍筋内部。

(b) Hook shall engage longitudinal reinforcement and the extension shall project into the interior of the stirrup or hoop.

6.5 受拉钢筋的锚固

6.5 Development of deformed bars in tension

受拉钢筋的锚固长度取下表计算值和305mm的最大值。

The development length of bars in tension shall be the greater of values in the following table and 305mm.

受拉钢筋的锚固长度与直径之比 l_d/d_b

The ratio of development length to diameter l_d/d_b

应用情况 Condition	fy(MPa)	≤No.19 fc'(MPa) 28	≤No.19 fc'(MPa) 35	≥No.22 fc'(MPa) 28	≥No.22 fc'(MPa) 35
符合(a)或(b)情况 Conform to(a)or(b)	280	25	23	32	28
	420	38	34	47	42
其它情况 Other	280	38	34	47	42
	420	57	51	71	64

(a) 钢筋在延伸或搭接处的净间距≥d_b，净保护层≥d_b，且箍筋在l_d范围内不小于规范要求的最小规定。

(a) Clear spacing of bars or wires being developed or lap spliced not less than d_b, clear cover at least d_b, and stirrups or ties throughout l_d not less than the Code minimum.

(b) 钢筋在延伸或搭接处的净间距≥2d_b，净保护层≥d_b。

(b) Clear spacing of bars or wires being developed or lap spliced at least 2d_b and clear cover at least d_b.

(c) 当纵筋下面的新浇筑混凝土的厚度大于305mm时，表中值应乘以1.3。

(c) When there are more than 305mm of fresh concrete placed below horizontal reinforcement, the ratio in table should be multiplied by 1.3.

6.6 受拉钢筋的弯钩锚固

6.6 Development of standard hooks in tension

受拉钢筋的弯钩锚固长度取下表计算值、8d_b和153mm的最大值。

The development length of bars in tension shall be the greater of values in the following table, 8d_b and 153mm。

受拉钢筋的弯钩锚固长度与直径之比 l_{dh}/d_b

The ratio of development fo standard hooks to diameter l_{dh}/d_b

应用情况 Cases	fy(MPa)	fc'(MPa) 28	fc'(MPa) 35
符合(a)或(b)情况 Conform to(a)or(b)	280	10	9
	420	15	14
其它情况 Other	280	13	11
	420	19	17

(a) 对于≤No.32带90度弯钩的钢筋，符合(1)沿l_{dh}段内的垂直封闭箍筋或拉筋间距s ≤ 3d_b或(2)沿l_{ext}段内的垂直封闭箍筋或拉筋间距s ≤ 3d_b。

(a) For 90-degree hooks of No. 32 and smaller bars

(1) enclosed along l_{dh} within ties or stirrups perpendicular to l_{dh} at s ≤ 3d_b or (2) enclosed along the bar extension beyond hook including the bend within ties or stirrups perpendicular to l_{ext} at s ≤ 3d_b.

(b) 对于≤No.32带180度弯钩的钢筋，符合沿l_{dh}段内的垂直封闭箍筋或拉筋间距s ≤ 3d_b。

(b) For 180-degree hooks of No. 32 and smaller bars enclosed along l_{dh} within ties or stirrups perpendicular to l_{dh} at s ≤ 3d_b.

6.7 受拉钢筋的搭接长度

6.7 Lap splice lengths of deformed bars in tension

A_{sp}/A_{sr}	搭接接头最大百分率 Maximum percent of A_s spliced within required lap length	搭接等级 Splice type	l_{st}
≥2.0	50	A级 Class A	$1.0l_d$和305mm取大值 Greater of $1.0l_d$ and 305mm
	100	B级 Class B	$1.3l_d$和305mm取大值 Greater of $1.3l_d$ and 305mm
<2.0	所有情况 All cases	B级 Class B	

A_{sp}/A_{sr}: 实际配筋面积与计算需要面积的比值

A_{sp}/A_{sr}: Ratio of area of reinforcement provided to area of reinforcement required by analysis at splice location

6.8 钢筋的机械及焊接连接

6.8 Mechanical and welded splices of deformed bars

6.8.1 钢筋采用机械或焊接连接的接头强度不得小于1.25f_y。

6.8.1 A mechanical or welded splice shall develop in tension or compression, as required, at least 1.25fy of the bar.

6.8.2 对受拉构件的钢筋应采用机械或焊接连接的接头，并应满足6.8.1条，相邻钢筋接头交错距离不得小于765mm。

6.8.2 Splices in tension tie members shall be made with a mechanical or welded splice in accordance with 6.8.1. Splices in adjacent tension bars shall be staggered at least 765mm.

6.8.3 钢筋焊接要求应符合 AWS D1.4 的规定。

6.8.3 Welding of all bars shall conform to the requirements of AWS D1.4.

7 施工图平法制图规则

7 Rules of ichnographic construction drawings

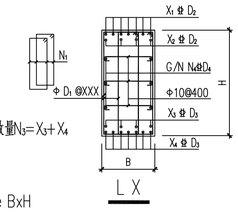
7.1 构件代码

7.1 Codes of the structural member

构件类型 Structural members	构件代码 Type code
柱，框架柱，梁上柱，暗柱，构造柱墙 Columns,Frame columns,Columns supported by the beam,Embedded columns,Constructional columns,Walls	Z，KZ，LZ，AZ，GZ,W
楼层框架梁，屋面框架梁，梁（非框架），悬挑梁，圈梁，暗梁 Floor frame beams,Roof frame beams,Beams (non-frame),Cantilever beams,Constructional beams, Embedded beams	KL，WKL，L，XL，QL,AL
基础承台，筏板基础，基础联系梁 Pile caps,Raft foundations,Connecting beams in foundations	CT，BPB，JLL

7.2 梁施工图平法制图规则

7.2 Rules of ichnographic construction drawings of beams



右图梁配筋截面的集中标注为：

(a)L XXX(X) BxH：梁L（跨数），编号XXX，截面尺寸BxH

(b)Φ D_1 @ XXX (N_1)：箍筋直径D_1，间距XXX，（肢数N_1）

(c)N_2 ⚤ D_2 X_1/X_2; N_3 ⚤ D_3 X_3/X_4：梁顶钢筋直径D2，数量$N_2=X_1+X_2$，梁底钢筋直径D_3，数量$N_3=X_3+X_4$

(d)G/N N_4 ⚤ D_4：腰筋G（构造）/N（扭筋）的直径D_4，数量N_4。

Concentrated annotation for beam sections in the right figure:

(a)L XXX(X) BxH：Beam L, No. XXX(No. of spans) and Section size BxH

(b)Φ D_1 @ XXX (N_1)：Stirrup diameter D_1, Space XXX and (Leg number)

(c)N_2 ⚤ D_2 X_1/X_2; N_3 ⚤ D_3 X_3/X_4: Longitudinal bars diameter D_2, Amount $N_2=X_1+X_2$ in beam top, Longitudinal bars diameter D_3, Amount $N_3=X_3+X_4$ in beam bottom

(d)G/N N_4 ⚤ D_4: Longitudinal bars diameter D_4 of construction G or torsion N, Amount N_4 in beam sides

8 图中未尽事宜按下列现行规程和标准执行，且不仅限于此。

8 The no mentioned matters in the drawings shall conform to the following codes which are not limited to these.

UBC-97-Uniform Building Code　　UBC-97-统一建筑规范

IBC-2015-International Building Code　　IBC-2015-国际建筑规范

ACI 318-14-Building Code Requirements for Structural Concrete　　ACI 318-14-混凝土建筑规范

ASTM A615/A615M-14-Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement　　ASTM A615/A615M-14-混凝土中螺纹钢及圆钢碳钢钢筋规范

ASTM A706/A706M-14-Standard Specification for Deformed and Plain Low-Alloy Steel Bars for Concrete Reinforcement　　ASTM A706/A706M-14-混凝土中螺纹钢及圆钢的低合金钢钢筋规范

AWS D1.1/D1.1M-2010-Structural Welding Code-Steel　　AWS D1.1/D1.1M-2010-钢材用结构焊接规范

AWS D1.4/D1.4M-2011-Structural Welding Code-Reinforcement　　AWS D1.4/D1.4M-2011-钢筋用结构焊接规范

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER TECHNOLOGY EQUIPMENT CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE：施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 8 0 1 - 0 0 1 A

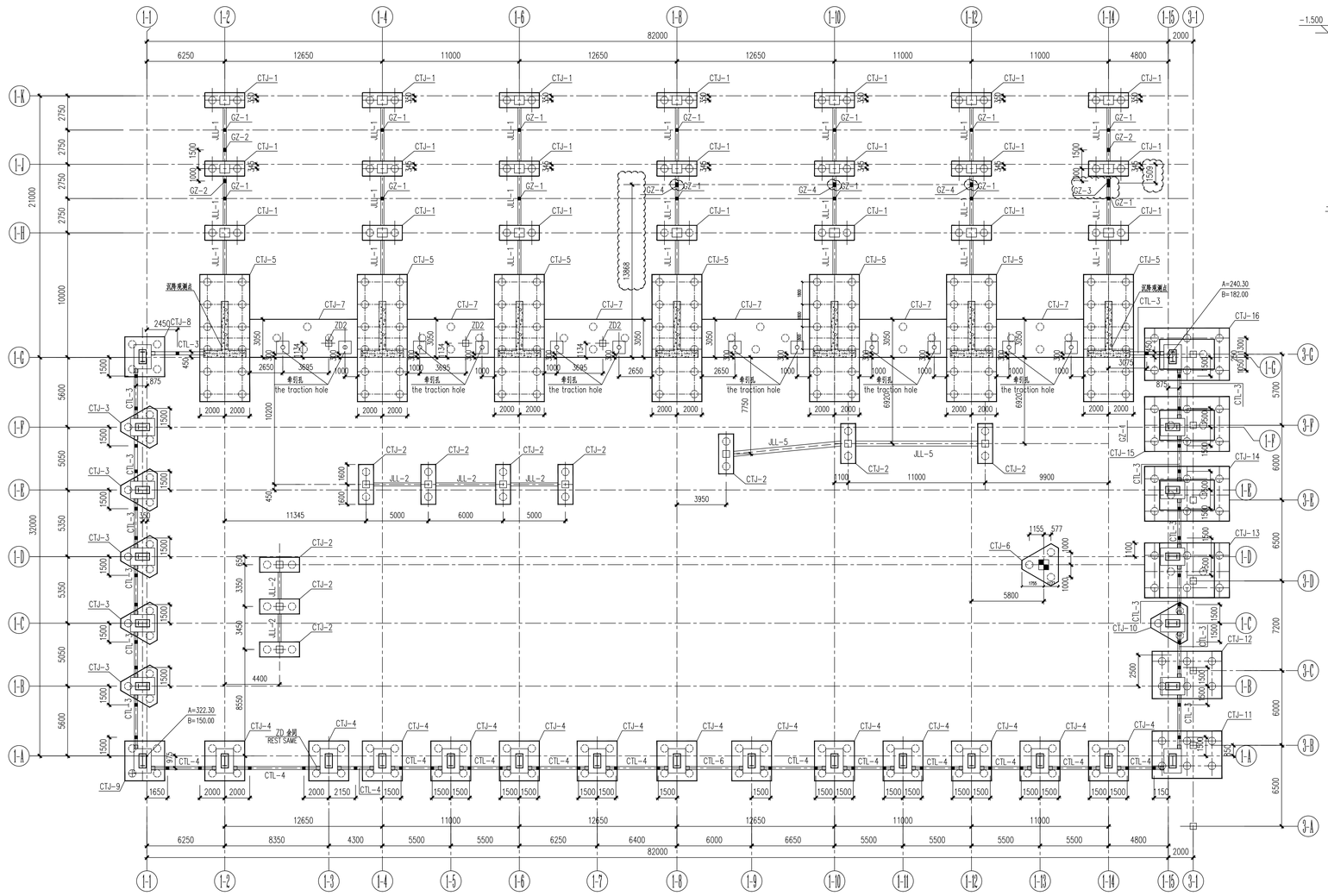
	姓名 Name	日期 Date
设计 Designed	王轩	2017.11.27
校核 Checked	金成博	2017.11.27
审核 Reviewed	王轩 刘红 赵刚	2017.11.27
批准 Approved	赵刚	2017.11.27

极1阀厅及防火墙基础施工图设计总说明

General specification of foundations for pole 1 valve hall and firewall

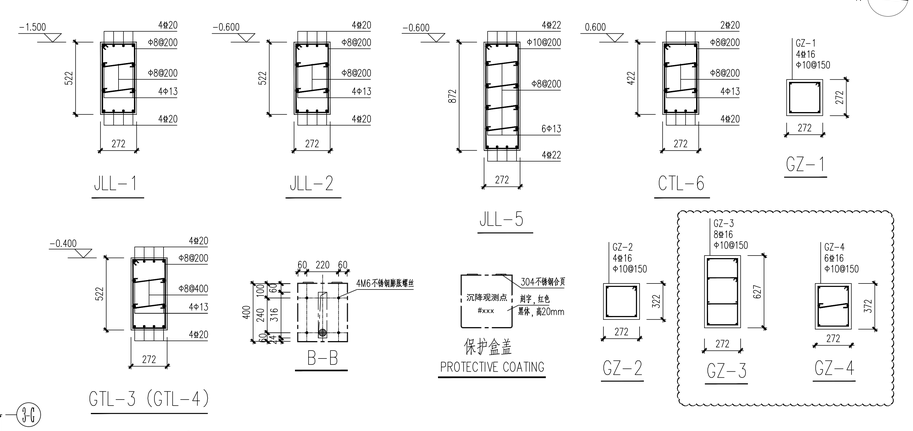
比例 Scale　　图号 DWG NO. 30-B003701S-T0801-001　　版本 REV. A

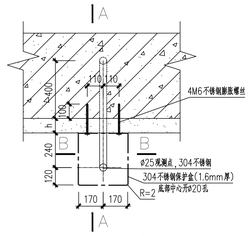
594X841



极1阀厅及防火墙基础承台布置图

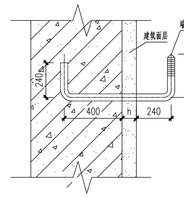
plan layout of pile cap for pole 1 valve hall and firewall

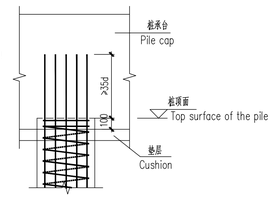




沉降观测点详图

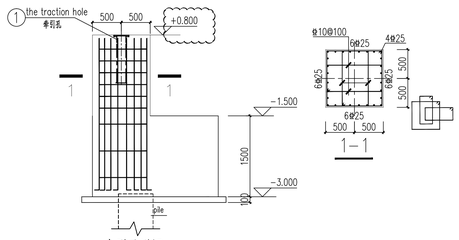
SETTLEMENT OBSERVATION POINT





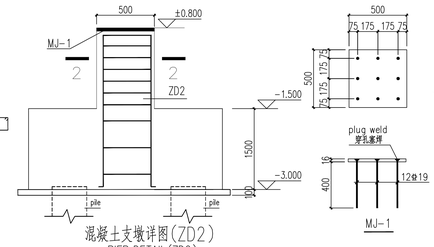
桩与承台连接构造

PILE-TOP LONGITUDINAL BAR'S ANCHORING DETAILING IN CAP



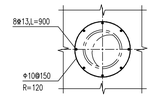
牵引孔详图

THE TRACTION HOLE



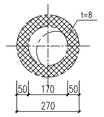
混凝土支墩详图(ZD2)

PIER DETAIL(ZD2)



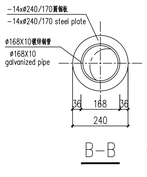
牵引孔四周加固钢筋

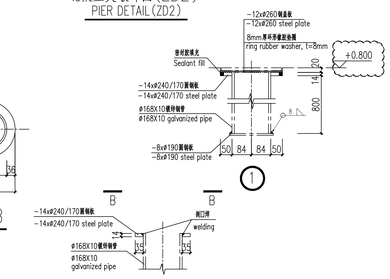
Reinforcement around the traction hole



橡胶垫圈大样

Rubber washer



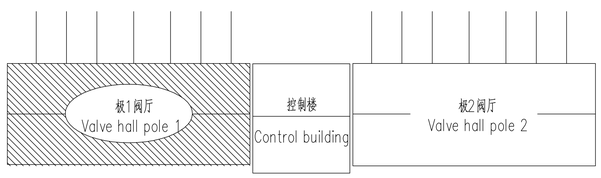


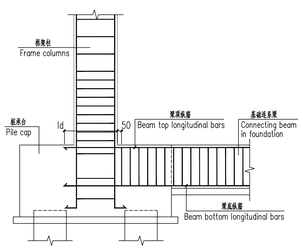
牵引孔预埋钢管大样

Embedded steel pipe of traction hole

1. 混凝土：基础、梁、柱：f'c=28MPa；垫层：f'c=21MPa。
2. 钢筋：Φ--ASTM A615 Grade 40，fy = 280MPa；
   Φ--ASTM A615 Grade 60，fy = 420MPa。
3. 最外层钢筋的混凝土保护层厚度：±0.000以下：基础、基础梁、柱：51mm；±0.000以上：框架柱、梁：40mm；构造柱：20mm。
4. 承台配筋详见04-06.
5. 未注明的钢筋锚固长度按01选用.
6. ■ 表示GZ-1.
7. 钢板、角钢和扁钢采用Q235B。
8. 牵引孔钢材采用Q345B。

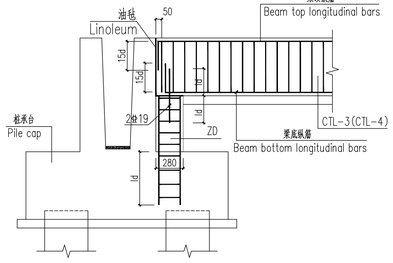
1. Concrete: for foundation, beam, column, roof, awning, parapet and air-condition set's foundation: fc'=28MPa; for cushion: fc'=21MPa.
2. Reinforcement:
   Φ--ASTM A615 Grade 40, fy = 280MPa;
   Φ--ASTM A615 Grade 60, fy = 420MPa.
3. Thickness of concrete cover for outermost rebar: for foundation, foundation beam and column that under ±0.000: 51mm; above ±0.000: for frame column and beam: 40mm; for tie column: 20mm.
4. Detail drawing of pile cap in 04-06.
5. The anchorage length is not marked should be located in 01.
6. ■represent GZ-1.
7. The structural carbon steel , grade Q235B, should be used for steel plate, angle steel and steel strip.
8. The structural carbon steel , grade Q345B, should be used for the traction hole.





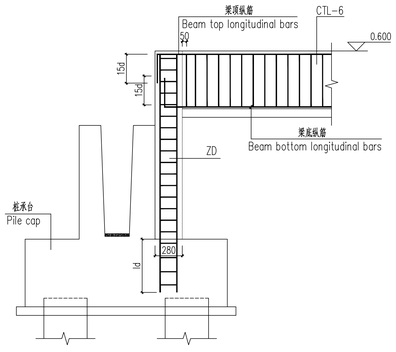
基础连系梁JLL配筋构造

REINFORCEMENT DETAILING OF CONNECTING BEAM IN FOUNDATION JLL (I)



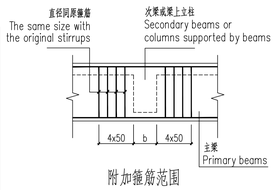
支墩ZD配筋构造(1)

REINFORCEMENT DETAILING OF SUPPORT PIERS(1)



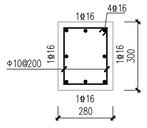
支墩ZD配筋构造(2)

REINFORCEMENT DETAILING OF SUPPORT PIERS(2)



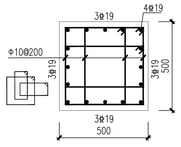
附加箍筋范围

RANGE OF ADDITIONAL STIRRUPS



混凝土支墩详图(ZD)

PIER DETAIL(ZD)



2—2

混凝土支墩详图(ZD2)

PIER DETAIL(ZD2)

Reference Drawings

Drwg. No.	Title
30-B003701S-T0401-001	Plan layout of foundation for pole 1 converter transformer
30-B003701S-T0401-002	Plan layout of foundation for pole 2 converter transformer
30-B003701S-T0202-002	Plan layout of foundations for the control building
30-B003701S-T0202-005	Plan layout of connecting beams in foundation for the control building

Revisions				
A	王轩	2017.11.15	First issue	
	Name	Date	Notes	

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 State Grid 中国南方电网有限责任公司 China Southern Power Grid Co., Ltd.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 8 0 1 - 0 0 2 A

Speciality	Sign	Date		Name	Date
电气	石建辉	2017.11.27	设计 Designed	王轩	2017.11.27
建筑	崔清民	2017.11.27	校核 Checked	安洪涛	2017.11.27
总图	赵东晖	2017.11.27	审核 Reviewed	王学刚	2017.11.27
			批准 Approved		2017.11.27

比例 Scale

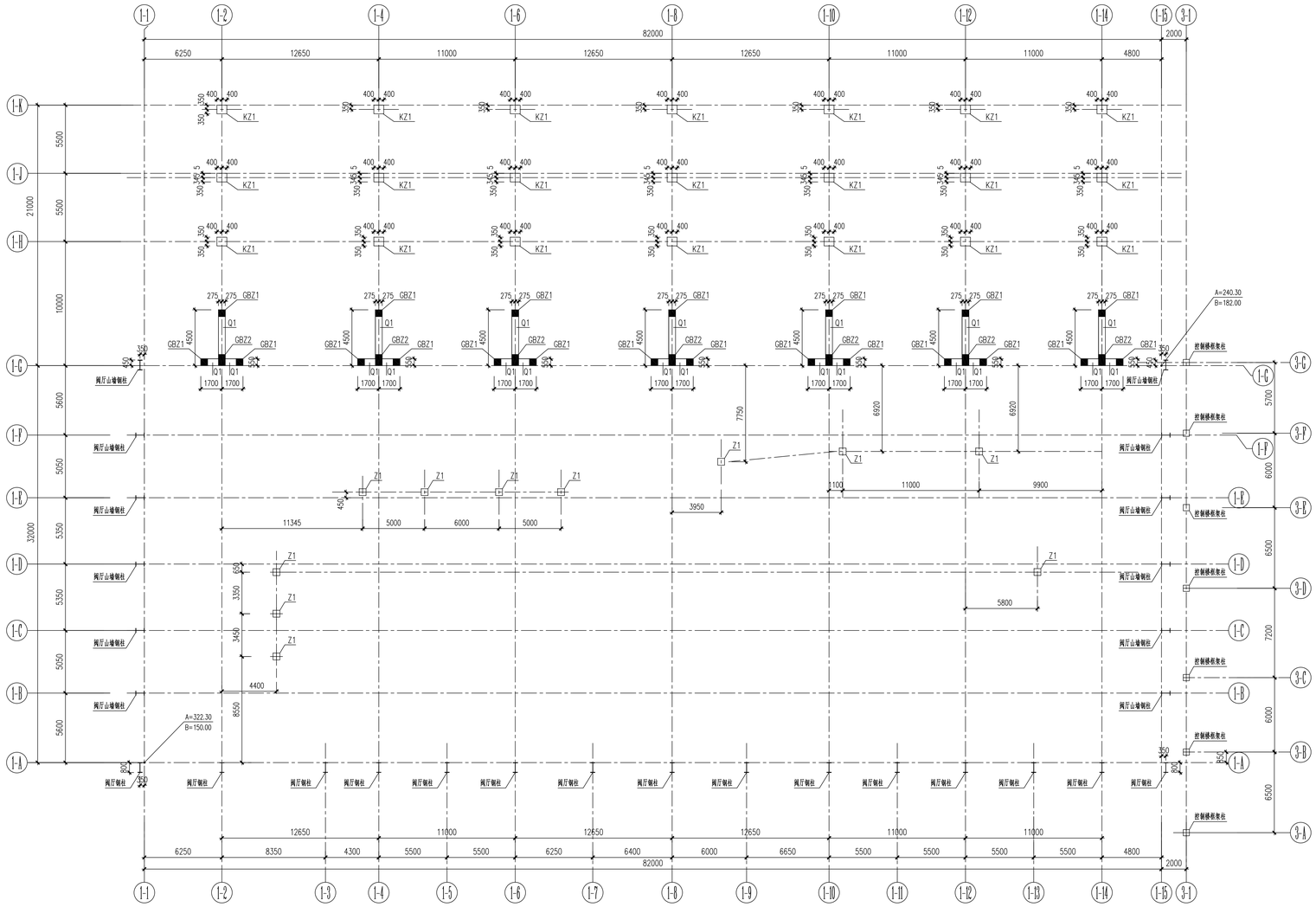
图号 DWG NO. 30-B003701S-T0801-002

版本 REV. A

极1阀厅及防火墙承台布置图

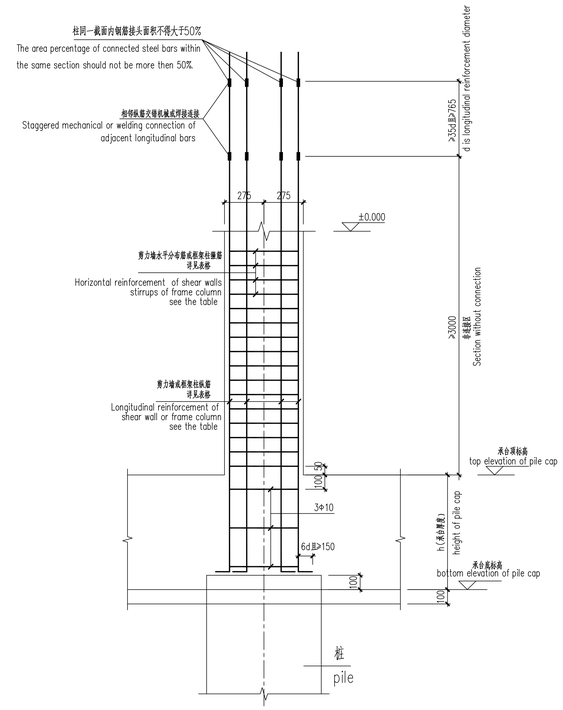
plan layout of pile cap for pole 1 valve hall and firewall

841X1189



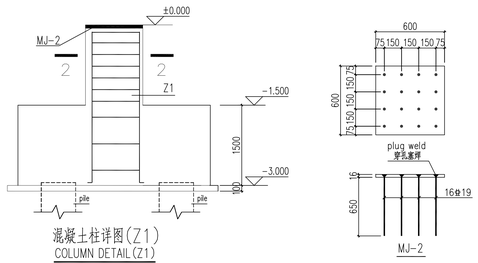
极1阀厅±0.000m结构布置图

±0.000m structure layout for pole 1 valve hall and firewall



基础预留剪力墙和框架柱插筋图

DETAILING OF WALL AND COLUMN'S INSERTED REINFORCEMENT IN FOUNDATION



混凝土柱详图(Z1)

COLUMN DETAIL(Z1)

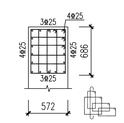
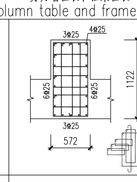
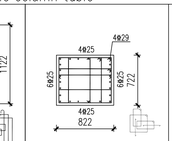
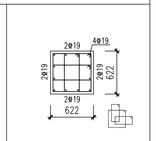
剪力墙身表

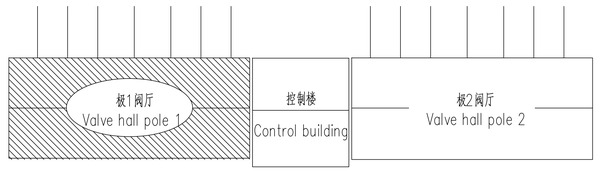
Shear wall table

编号 NO	标高 elevation	墙厚 thickness	水平分布筋(三层) Horizontal distributed reinforcement(three layer)	竖向分布筋(三层) Vertical distribution bars (three layer)	拉筋(双向) binding rebar (two way)
Q1	-1.500~±0.000	550	⌀16@150(3)	⌀16@150(3)	Φ10@450/@450

剪力墙柱表和框架柱表

Shear wall column table and frame column table

截面 section				
编号 NO	GBZ1	GBZ2	KZ1	Z1
标高 elevation	承台顶标高~±0.000	承台顶标高~±0.000	承台顶标高~±0.000	承台顶标高~±0.000
纵筋 Longitudinal reinforcement	18⌀25	22⌀25	4⌀29+20⌀25	12⌀19
箍筋 stirrup	⌀13@100	⌀13@100	⌀13@100	Φ10@200



1.钢板、角钢和扁钢采用Q235B。

1.The structural carbon steel , grade Q235B, should be used for steel plate, angle steel and steel strip.

Reference Drawings

Drwg. No.	Title
30-B003701S-T0401-001	Plan layout of foundation for pole 1 converter transformer
30-B003701S-T0401-002	Plan layout of foundation for pole 2 converter transformer
30-B003701S-T0202-002	Plan layout of foundations for the control building
30-B003701S-T0202-003	Plan layout of connecting beams in foundation for the control building

Revisions

	Name	Date	Notes
A	王轩	2017.11.15	First issue

巴基斯坦联营至里-拉合尔±660kV直流输电工程

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 8 0 1 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	王轩	2017.11.27
校核 Checked	金武峰	2017.11.27
审核 Reviewed	王轩 刘利利	2017.11.27
批准 Approved		2017.11.27

极1阀厅±0.000m结构布置图

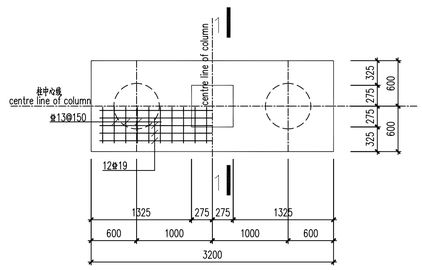
±0.000m structure layout for pole 1 valve hall and firewall

比例 Scale

图号 DWG. NO. 30-B003701S-T0801-003

版本 REV. A

841X1189



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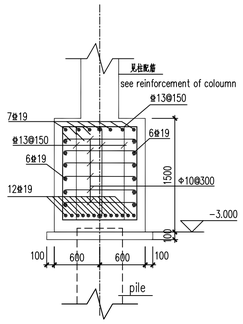

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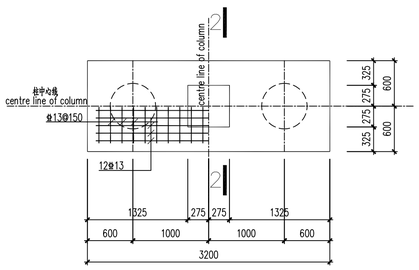

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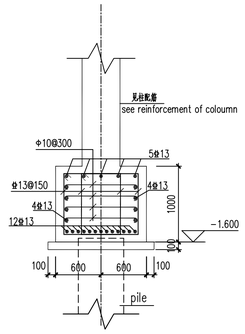

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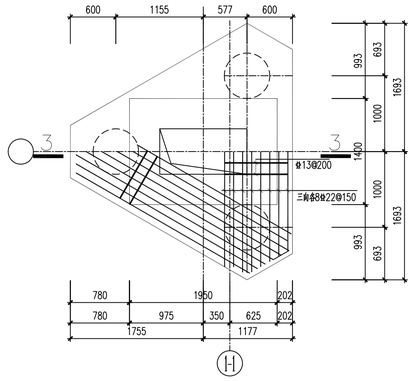

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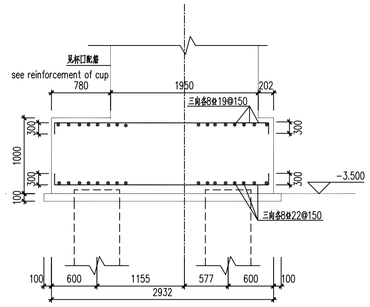

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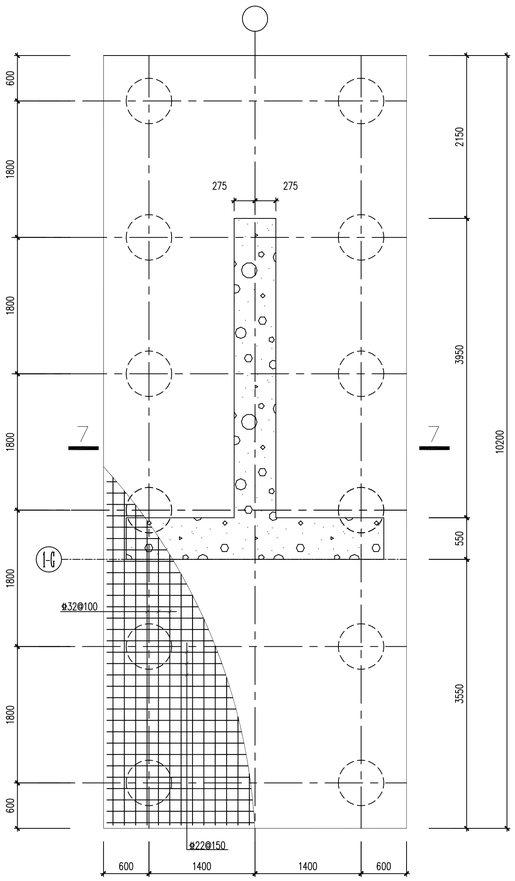

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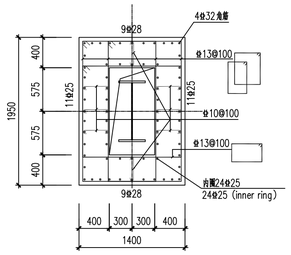

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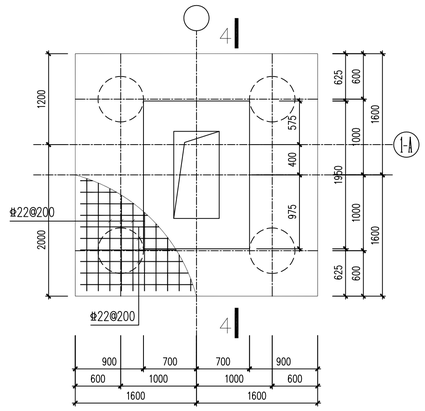

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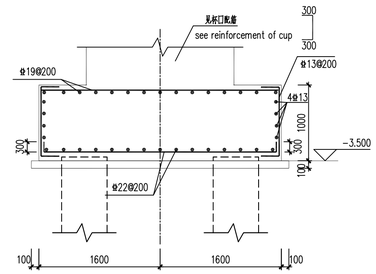

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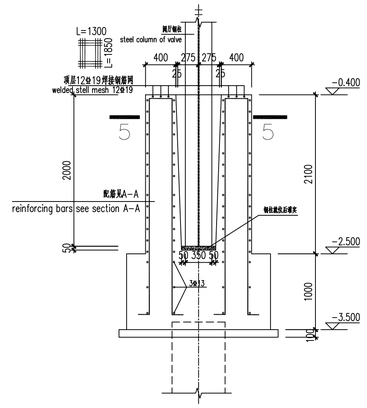
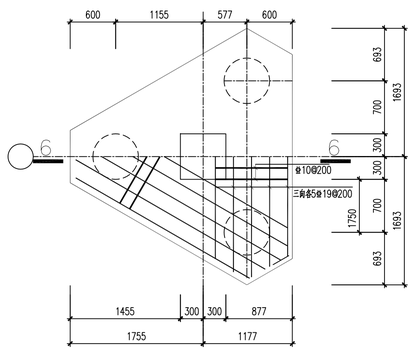


杯口配筋
CUP RABBET



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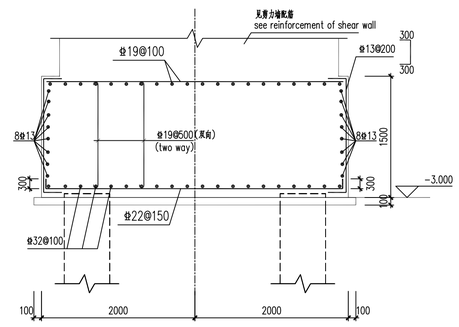

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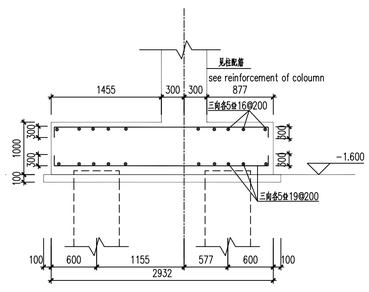

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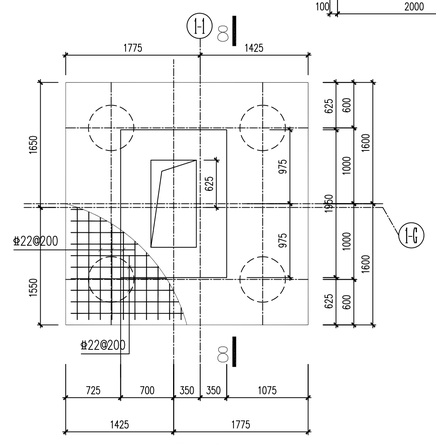
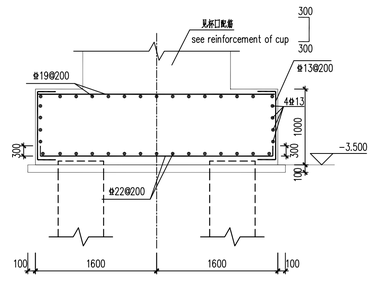


CTJ-8



8-8

1. 未注明的钢筋锚固长度按01选用.

1. The anchorage length is not marked should be located in 01.

Reference Drawings

Drwg. No.	Title
30-B003701S-T0401-001	Plan layout of foundation for pole 1 converter transformer
30-B003701S-T0401-002	Plan layout of foundation for pole 2 converter transformer
30-B003701S-T0202-002	Plan layout of foundations for the control building
30-B003701S-T0202-005	Plan layout of connecting beams in foundation for the control building

Revisions

	Name	Date	Notes
A	王轩	2017.11.15	First issue

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 8 0 1 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	王轩	2017.11.27
校核 Checked	金成峰	2017.11.27
审核 Reviewed	郑明祖	2017.11.27
批准 Approved	赵刚	2017.11.27
比例 Scale		

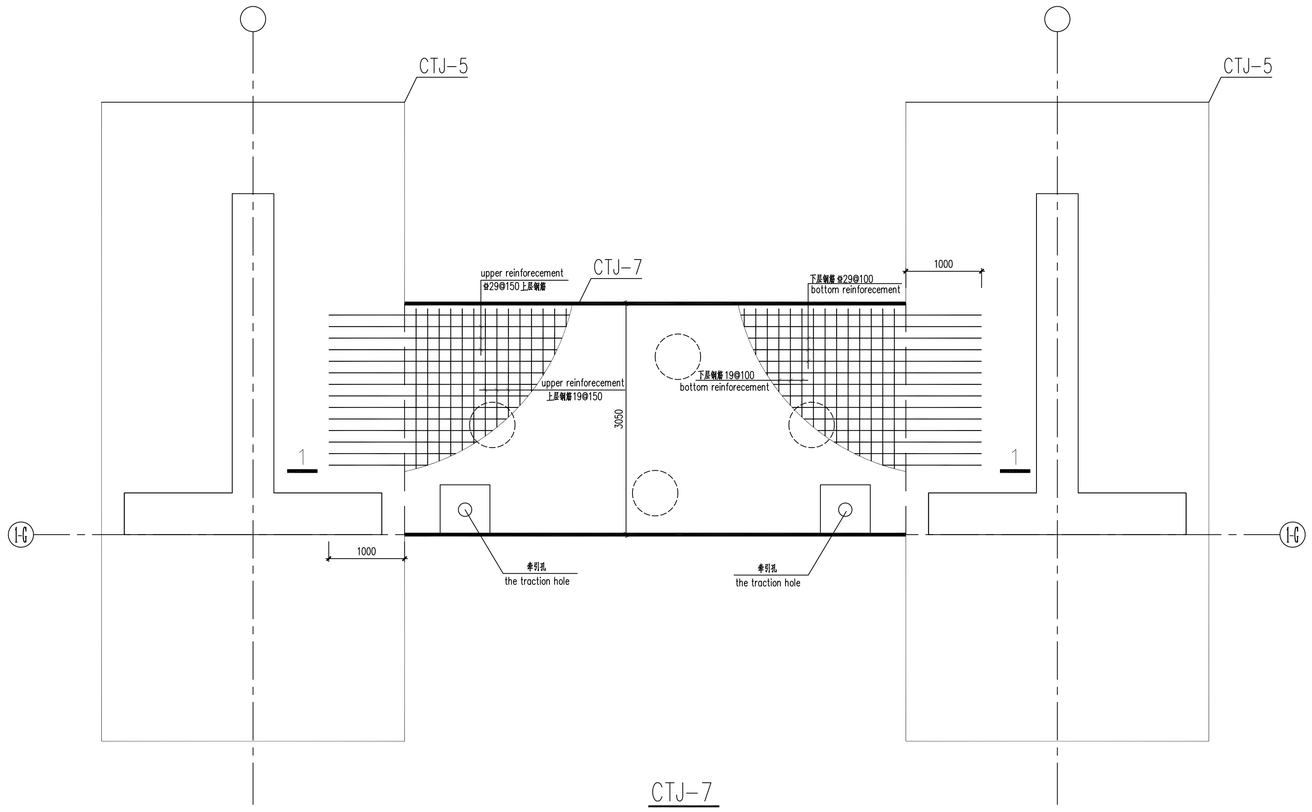
极1阀厅及防火墙承台详图(1)

Detail drawing of pile cap for pole 1 valve hall and firewall(1)

图号 DWG NO.: 30-B003701S-T0801-004

版本 REV.: A

594X841



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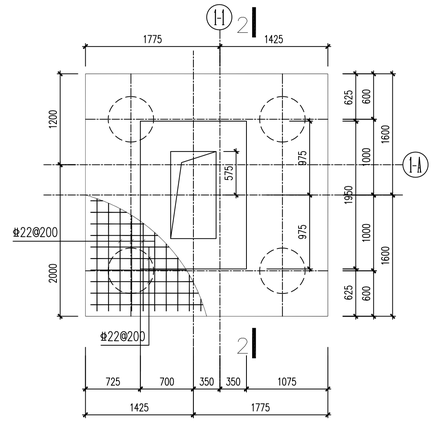

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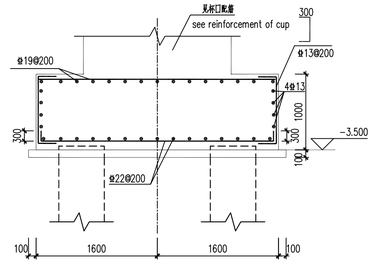

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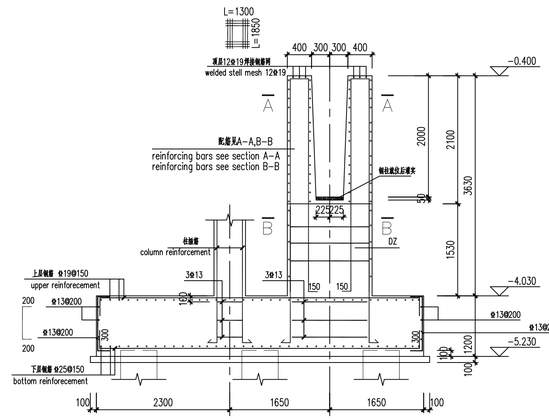

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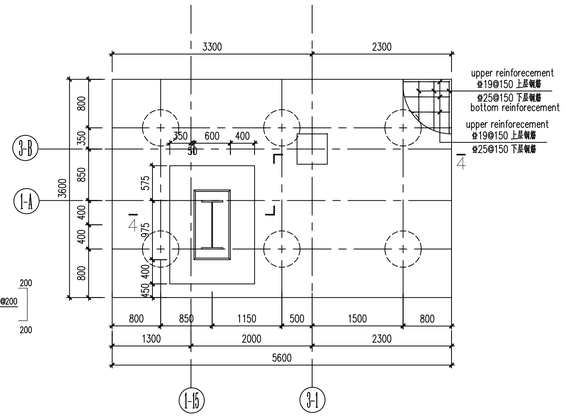

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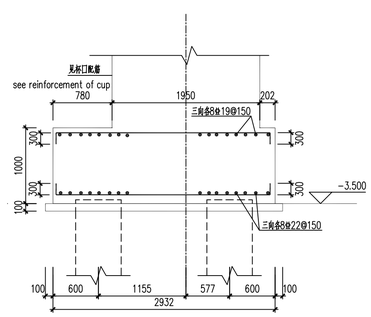

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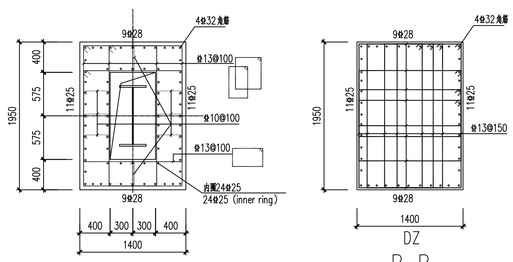

A-A

B-B

1. 未注明的钢筋锚固长度按01选用.

1. The anchorage length is not marked should be located in 01.

Reference Drawings

Drwg. No.	Title
30-B003701S-T0401-001	Plan layout of foundation for pole 1 converter transformer
30-B003701S-T0401-002	Plan layout of foundation for pole 2 converter transformer
30-B003701S-T0202-002	Plan layout of foundations for the control building
30-B003701S-T0202-005	Plan layout of connecting beams in foundation for the control building

Revisions

A	王轩	2017.11.15	First issue
	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 8 0 1 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	王轩	2017.11.27
校核 Checked	金沃源	2017.11.27
审核 Reviewed	李卫 刘利 赵刚	2017.11.27
批准 Approved	刘利	2017.11.27
比例 Scale		

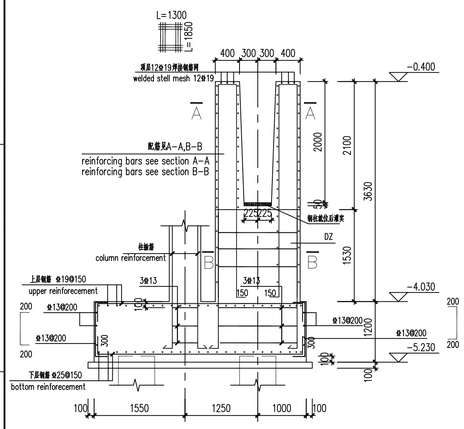
极1阀厅及防火墙承台详图(2)

Detail drawing of pile cap for pole 1 valve hall and firewall(2)

图号 DWG NO. 30-B003701S-T0801-005

版本 REV. A

594X841



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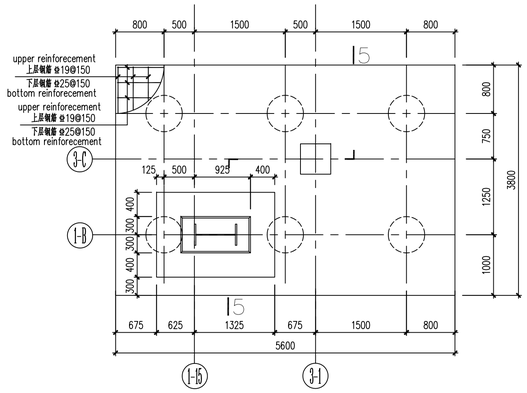

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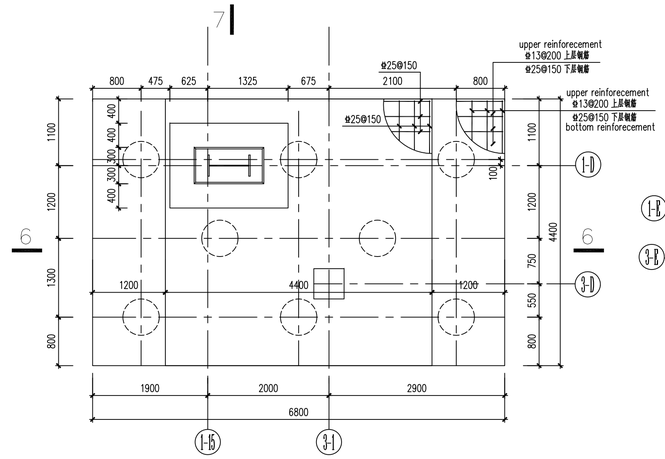

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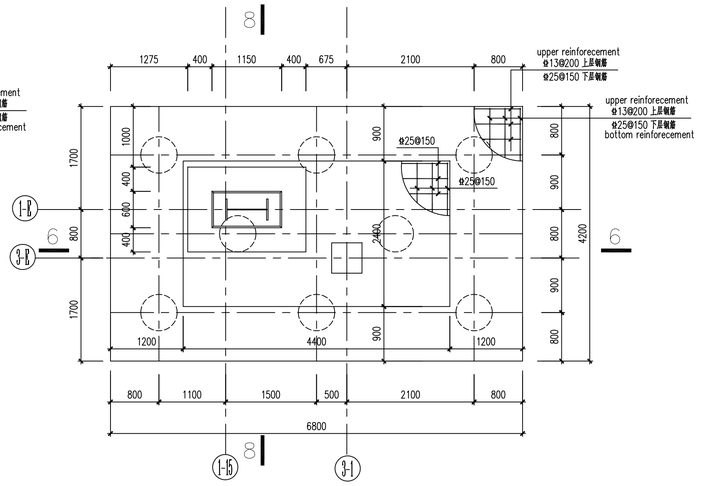

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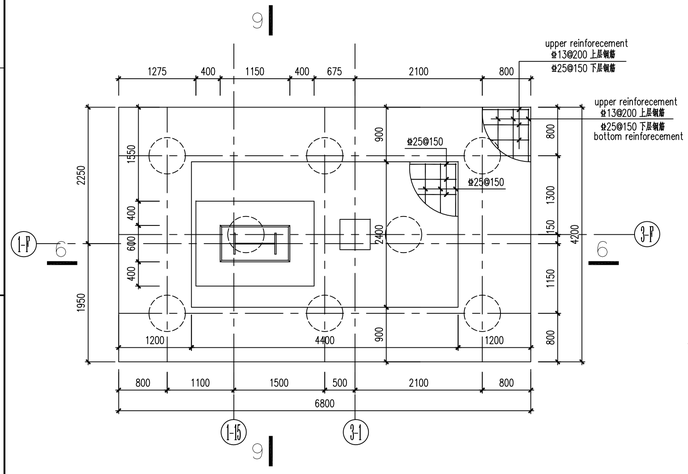

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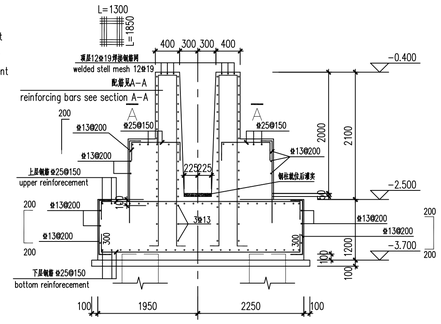

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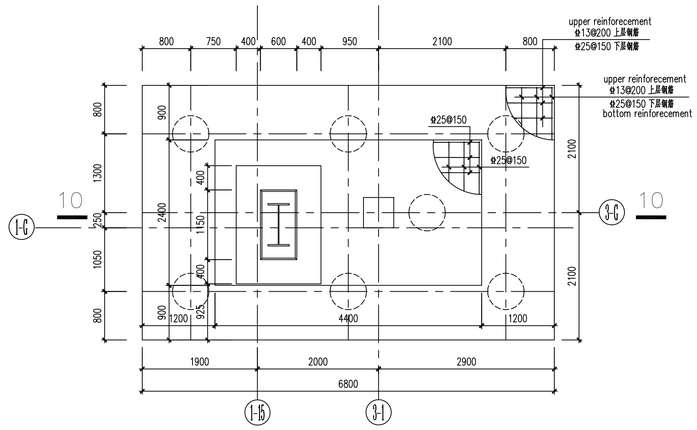

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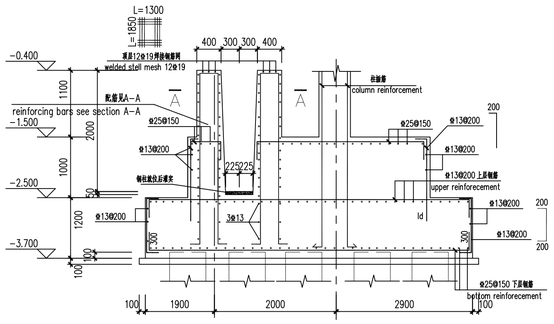
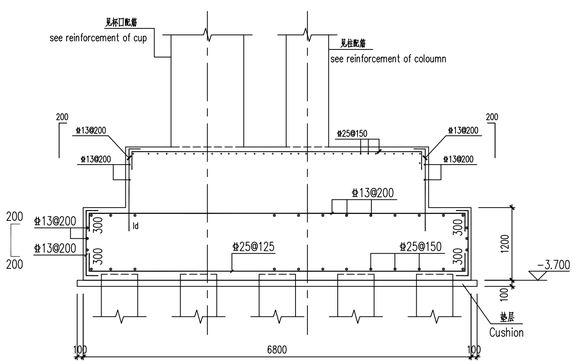


10-10

1. 未注明的钢筋锚固长度按01选用.

1. The anchorage length is not marked should be located in 01.



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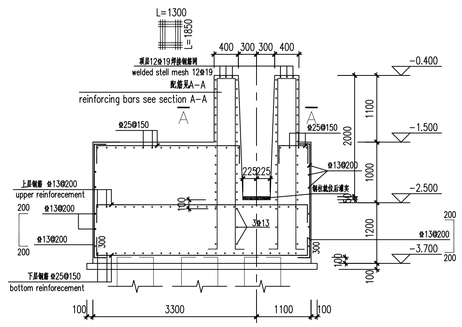

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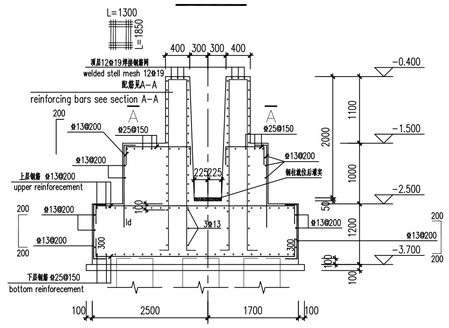

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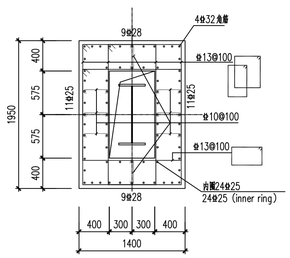
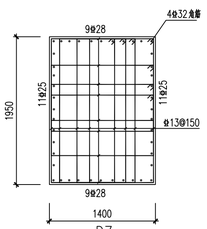


A-A



B-B

Reference Drawings

Drwg. No.	Title
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30-B003701S-T0401-002	Plan layout of foundation for pole 2 converter transformer
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Revisions			
A	王轩	2017.11.15	First issue
	Name	Date	Notes

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 8 0 1 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	王轩	2017.11.27
校核 Checked	金成源	2017.11.27
审核 Reviewed	王印刚	2017.11.27
批准 Approved	[illegible]	2017.11.27
比例 Scale		

极1阀厅及防火墙承台详图(3)

Detail drawing of pile cap for pole 1 valve hall and firewall(3)

图号 DWG NO.: 30-B003701S-T0801-006

版本 REV.: A

594X841